

Lecture 2

Probability Essentials

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(Edited from the lecture notes by Brian J. Reich, NC State)



Review of probability (Recap)

- ▶ In Frequentist or classical statistics, we assume that the data $\mathbf{Y}$ follows a model that involves some unknown fixed parameter θ .
- ▶ In Bayesian statistics, we also assume a probability distribution (prior) for θ .
- ▶ The most important thing in Bayesian statistics is computing the posterior, i.e., the distribution of the parameters θ after observing the data $\mathbf{Y}$.
- ▶ This is the conditional distribution of θ given $\mathbf{Y}$.
- ▶ Therefore, we need to review the probability concepts that lead to the conditional distribution of one variable conditioned on another.
 1. Probability mass (PMF) and density (PDF) functions (Lecture 1)
 2. Joint distributions (Lecture 2)
 3. Marginal and conditional distributions (Lecture 2)

Joint distributions

- ▶ $\mathbf{X} = (X_1, \dots, X_p)$ is a random vector (vectors and matrices should be in bold).
- ▶ For convenience, let's consider only $p = 2$ random variables X and Y .
- ▶ (X, Y) is discrete if it can take on a countable number of values, such as
 X = number of students who were tested for Covid at IIT-K and
 Y = number of students who tested positive.
- ▶ (X, Y) is continuous if it can take on an uncountable number of values, such as
 X = height of a person and Y = weight of a person.

Discrete random variables

- ▶ The joint PMF is

$$f(x, y) = \text{Prob}(X = x, Y = y)$$

- ▶ Example: patients are randomly assigned a dose and followed to determine whether they develop a tumor.
- ▶ $X \in \{10, 20, 30\}$ is the dose; $Y \in \{0, 1\}$ is 1 if a tumor develops and 0 otherwise
- ▶ The joint PMF is

	X		
Y	10	20	30
0	0.469	0.124	0.049
1	0.231	0.076	0.051

Discrete random variables

- ▶ The **marginal PMF** for X is

$$f_X(x) = \text{Prob}(X = x) = \sum_y f(x, y).$$

- ▶ The **marginal PMF** for Y is

$$f_Y(y) = \text{Prob}(Y = y) = \sum_x f(x, y).$$

- ▶ The marginal distribution is the same as univariate distribution as if we ignored the other variable.

Discrete random variables

- ▶ Example: X = dose and Y = tumor status.
- ▶ Find the marginal PMFs of X and Y .
- ▶ The joint PMF:

Y	X		
	10	20	30
0	0.469	0.124	0.049
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Discrete random variables

- ▶ Example: X = dose and Y = tumor status.
- ▶ Find the marginal PMFs of X and Y .
- ▶ The joint PMF:

Y	X		
	10	20	30
0	0.469	0.124	0.049
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$$\begin{aligned} & \text{Prob}(Y = 0) \\ = & \text{Prob}(X = 10, Y = 0) + \text{Prob}(X = 20, Y = 0) + \text{Prob}(X = 30, Y = 0) \\ = & 0.469 + 0.124 + 0.049. \end{aligned}$$

Discrete random variables

- ▶ The **Conditional PMF** of Y given X is

$$f(y|x) = \text{Prob}(Y = y|X = x) = \frac{\text{Prob}(X = x, Y = y)}{\text{Prob}(X = x)} = \frac{f(x, y)}{f_X(x)}.$$

- ▶ Here x is treated as a fixed number, and so $f(y, x)$ is only a function of y .
- ▶ However, we can't use $f(x, y)$ as the PMF for Y because

$$\sum_y f(x, y) = f_X(x) \neq 1$$

- ▶ Dividing by $f_X(x)$ makes $f(y|x)$ valid

$$\sum_y f(y|x) = \sum_y \frac{f(y, x)}{f_X(x)} = \frac{\sum_y f(y, x)}{f_X(x)} = \frac{f_X(x)}{f_X(x)} = 1$$

Discrete random variables

- ▶ Example: X = dose and Y = tumor status.
- ▶ Find $f(y|x)$.
- ▶ The joint PMF:

Y	X		
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Discrete random variables

- ▶ Example: X = dose and Y = tumor status.
- ▶ Find $f(y|x)$.
- ▶ The joint PMF:

Y	X		
	10	20	30
0	0.469	0.124	0.049
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$$\begin{aligned} & \text{Prob}(Y = 0|X = 10) \\ &= \text{Prob}(X = 10, Y = 0)/\text{Prob}(X = 10) \\ &= 0.469/(0.469 + 0.231). \end{aligned}$$

Monte Hall problem

- ▶ https://en.wikipedia.org/wiki/Monty_Hall_problem
- ▶ Suppose you're on a game show, and you're given the choice of three doors.
- ▶ Behind one door is a car; behind the others, goats.
- ▶ You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat.
- ▶ He then says to you, "Do you want to pick door No. 2?"
- ▶ **Is it to your advantage to switch your choice?**

Monte Hall problem

Door 1	Door 2	Door 3	Staying at Door 1	Switching to the door offered
Goat	Goat	Car	Wins goat	Wins car
Goat	Car	Goat	Wins goat	Wins car
Car	Goat	Goat	Wins car	Wins goat

Discrete random variables

- ▶ X and Y are **independent** if

$$f(x, y) = f_X(x)f_Y(y)$$

for all x and y .

- ▶ Variables are dependent if they are not independent.
- ▶ Equivalently, X and Y are **independent** if

$$f(x|y) = f_X(x)$$

for all x and y .

- ▶ These two definitions are equivalent.

$$f(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{f_X(x)f_Y(y)}{f_Y(y)} = f_X(x).$$

Discrete random variables

- ▶ Notation: $X_1, \dots, X_n \stackrel{iid}{\sim} f(x)$ means that $X_1, \dots, X_n$ are independent and identically distributed.

- ▶ This implies the joint PMF is

$$\text{Prob}(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n f(x_i).$$

- ▶ The same notation and definitions of independence apply to continuous random variables.
- ▶ In this class, assume independence unless otherwise noted.

Hurricane proportions by landfall (X) and category (Y)

	Category					Total
	1	2	3	4	5	
US (1)	0.0972	0.0903	0.0694	0.0069	0.0069	0.2708
Not US (0)	0.3194	0.1319	0.1389	0.1181	0.0208	0.7292
Total	0.4167	0.2222	0.2083	0.1250	0.0278	1.0000

Problem: Prove X and Y are dependent.

Hurricane proportions by landfall (X) and category (Y)

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Problem: Prove X and Y are dependent.

► $\text{Prob}(X = 1) = 0.2708$, $\text{Prob}(Y = 1) = 0.4167$.

So, $\text{Prob}(X = 1) \times \text{Prob}(Y = 1) = 0.2708 \times 0.4167 = 0.1128$.

Thus, $\text{Prob}(X = 1) \times \text{Prob}(Y = 1) \neq \text{Prob}(X = 1, Y = 1)$.

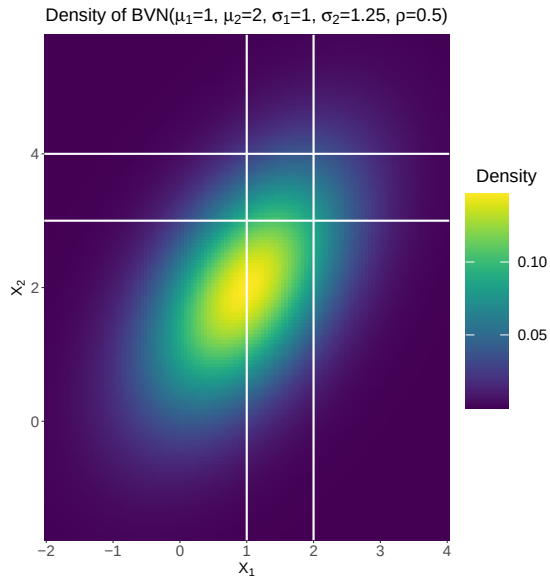
Continuous random variables

- ▶ Manipulating joint PDFs is similar to joint PMFs but sums are replaced by integrals.
- ▶ The joint PDF is denoted $f(x, y)$.
- ▶ Probabilities are computed as volume under the PDF:

$$\text{Prob}[(X, Y) \in A] = \int_A f(x, y) dx dy$$

where $A \subset \mathbb{R}^2$.

Continuous random variables



Continuous random variables

- ▶ Example: X =birthweight, Y =gestational age.
- ▶ Domain: $X \in (2, 10)$ lbs and $Y \in (20, 50)$ weeks.
- ▶ PDF: $f(x, y) = 0.26 \exp(-|x - 7| - |y - 40|)$.
- ▶ Find: $\text{Prob}(X > 7, Y > 40)$.

Continuous random variables

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- ▶ Find: $\text{Prob}(X > 7, Y > 40)$.
- ▶

$$\begin{aligned} & \text{Prob}(X > 7, Y > 40) \\ &= \int_7^{10} \int_{40}^{50} 0.26 \exp(-|x - 7| - |y - 40|) dx dy \\ &= 0.26 \left(\int_0^3 \exp(-x) dx \right) \times \left(\int_0^{10} \exp(-y) dy \right). \end{aligned}$$

Continuous random variables

- ▶ The **Marginal PDF** of X is

$$f_X(x) = \int f(x, y) dy.$$

- ▶ f_X is the univariate PDF for X as if we never considered Y .
- ▶ Find: $f_X(x)$ for the birthweight example.

Continuous random variables

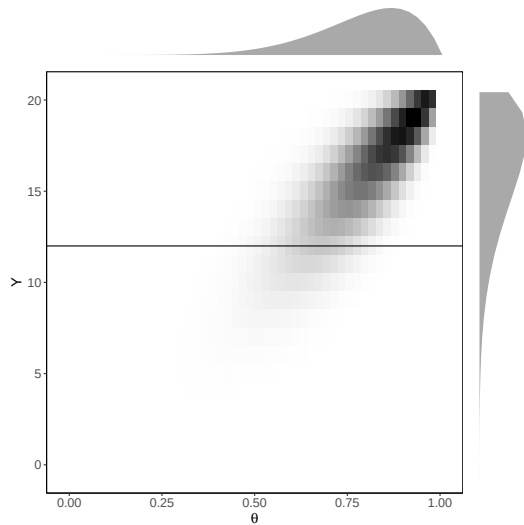
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- ▶ f_X is the univariate PDF for X as if we never considered Y .
- ▶ Find: $f_X(x)$ for the birthweight example.
- ▶

$$\begin{aligned} f_X(x) &= \int_{20}^{50} 0.26 \exp(-|x - 7| - |y - 40|) dy \\ &= 0.26 \exp(-|x - 7|) \times \int_{20}^{50} \exp(-|y - 40|) dy. \end{aligned}$$

Joint and marginal distributions



Continuous random variables

- ▶ The **Conditional PDF** of Y given X is

$$f(y|x) = \frac{f(x, y)}{f_X(x)}.$$

- ▶ Proper: $\int f(y|x) dy = \int \frac{f(x, y)}{f_X(x)} dy = \frac{\int f(x, y) dy}{f_X(x)} = 1.$
- ▶ Find: $f(y|x)$ for the birthweight example.

Continuous random variables

- ▶ The **Conditional PDF** of Y given X is

$$f(y|x) = \frac{f(x, y)}{f_X(x)}.$$

- ▶ Proper: $\int f(y|x) dy = \int \frac{f(x, y)}{f_X(x)} dy = \frac{\int f(x, y) dy}{f_X(x)} = 1.$

- ▶ Find: $f(y|x)$ for the birthweight example.



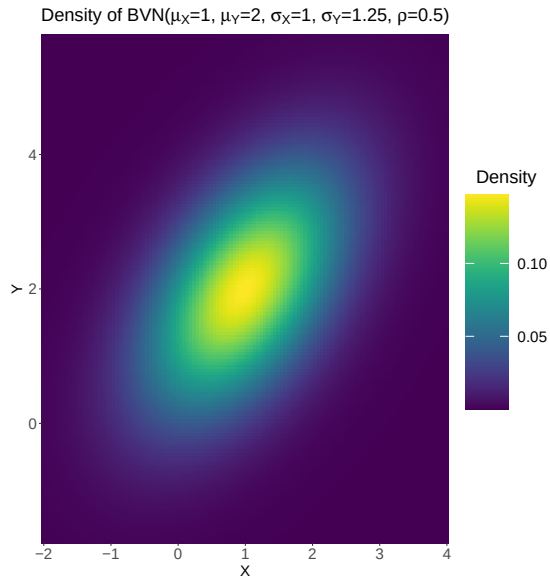
$$\begin{aligned} f(y|x) &= \frac{f(x, y)}{f_X(x)} \\ &= 0.26 \exp(-|x - 7| - |y - 40|) / \int_{20}^{50} 0.26 \exp(-|x - 7| - |y - 40|) dy \\ &= \exp(-|y - 40|) / \int_{20}^{50} \exp(-|y - 40|) dy. \end{aligned}$$

Bivariate normal distribution

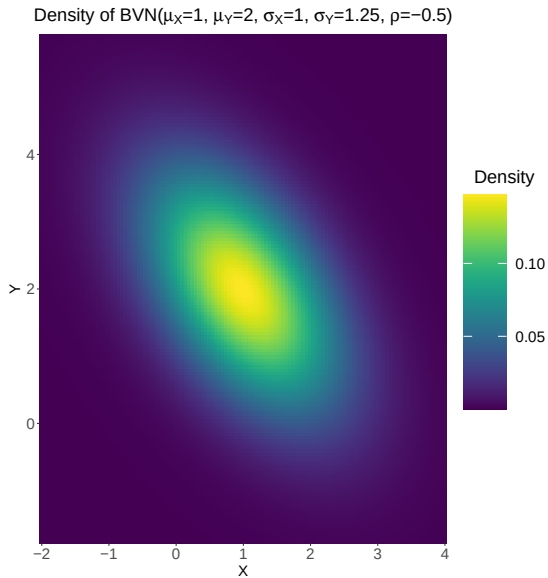
- ▶ The **bivariate normal distribution (BVN)** is the most common multivariate family.
- ▶ There are 5 parameters:
 - ▶ The marginal means of X and Y are μ_X and μ_Y .
 - ▶ The marginal variances of X and Y are $\sigma_X^2 > 0$ and $\sigma_Y^2 > 0$.
 - ▶ The correlation between X and Y is $\rho \in (-1, 1)$.
- ▶ The joint PDF is $f(x, y) =$

$$\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \left\{ -\frac{\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)}{2(1-\rho^2)} \right\}$$

Bivariate normal distribution



Bivariate normal distribution



Bivariate normal distribution

- ▶ The marginal distributions are $X \sim \text{Normal}(\mu_X, \sigma_X^2)$ and $Y \sim \text{Normal}(\mu_Y, \sigma_Y^2)$
- ▶ The conditional distribution is

$$Y|X \sim \text{Normal} \left\{ \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (X - \mu_X), \sigma_Y^2 (1 - \rho^2) \right\}$$

- ▶ The standard BVN distribution has $\mu_X = \mu_Y = 0$ and $\sigma_X = \sigma_Y = 1$ and thus

$$Y|X \sim \text{Normal}(\rho X, 1 - \rho^2)$$

- ▶ Explain the role of ρ .
- ▶ See “BVN marginal” and “BVN conditional” in the derivations uploaded at Hello IIT-K.

Defining joint distributions conditionally

- ▶ Specifying joint distributions is hard.
- ▶ Every joint distribution can be written

$$f(x, y) = f(y|x)f(x).$$

- ▶ Therefore, any joint distribution can be defined by
 1. X 's marginal distribution,
 2. The conditional distribution of $Y|X$.
- ▶ The joint problem reduces to two univariate problems.
- ▶ This idea forms the basis of hierarchical modeling.

Defining joint distributions conditionally

- ▶ Let X be the number of robins in the forest.
- ▶ Let Y be the number of robins we observe.
- ▶ Model $\text{Prob}(X = x) = 1/20$ for $x \in \{0, \dots, 19\}$ and $Y|X \sim \text{Binomial}(X, 0.2)$.
- ▶ What is the support of (X, Y) ?
- ▶ What is $\text{Prob}(X = 10, Y = 1)$?
- ▶ What is $\text{Prob}(Y = 0)$?

Defining joint distributions conditionally

- ▶ $f(x, y)$ is positive over $x, y \in \{0, 1, \dots, 19\}$ such that $y \leq x$.



$$\begin{aligned} & \text{Prob}(X = 10, Y = 1) \\ &= \text{Prob}(X = 10) \times \text{Prob}(Y = 1|X = 10) \\ &= \frac{1}{20} \times \binom{10}{1} 0.2^1 0.8^9. \end{aligned}$$



$$\begin{aligned} \text{Prob}(Y = 0) &= \sum_{x=0}^{19} \text{Prob}(X = x, Y = 0) \\ &= \sum_{x=0}^{19} \text{Prob}(X = x) \times \text{Prob}(Y = 0|X = x) \\ &= \sum_{x=0}^{19} \frac{1}{20} \times \binom{x}{0} 0.2^0 0.8^x = \frac{1}{20} \times \sum_{x=0}^{19} 0.8^x. \end{aligned}$$